

Exact Forest Reduction for Sparse Provenance Completion: Matching Rank, One-Defect Closed Forms, and Dual Motif Packings

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Abstract

Sparse provenance completion reduces, component by component, to

$$\Gamma(b, C) = \min_{\theta \in \mathbb{C}^J} \|b + C\theta\|_1.$$

The preceding incidence decomposition identifies the exact nonzero bipartite graph but does not by itself solve a nontrivial component. We solve the forest-supported complex-scalar case to the extent permitted by its intrinsic defect.

First, a variable leaf and its adjacent analysis row can be erased without changing the value. Second, for a matrix with forest support and nonzero edge coefficients,

$$\text{rank}_{\mathbb{C}} C = \nu(G),$$

where $\nu(G)$ is the maximum matching number. Consequently a row-saturating matching is exactly the graph-theoretic condition for zero completion cost for every datum. After exhaustive variable-leaf erasure, every surviving variable has degree at least two, the variables are matchable, and each connected core satisfies

$$\dim_{\mathbb{C}} \text{Ker } C^* = |I| - |J| = 1 + \sum_{j \in J} (\deg(j) - 2).$$

Thus a core has one complex obstruction precisely when every surviving variable has degree two.

For any one-defect component, if $\text{Ker } C^* = \mathbb{C}\alpha$, we obtain the closed formula

$$\Gamma(b, C) = \frac{|\langle \alpha, b \rangle|}{\|\alpha\|_{\infty}},$$

together with explicit primal and dual optimizers and the complete optimal-residual face. On a degree-two tree, α is recovered by edgewise phase transport. For branching cores we give an exact matching-based reduction to a complex second-order cone program having only $\dim \text{Ker } C^*$ variables. We also prove an exact column-dominant pivot rule and additive or fractional lower bounds from balanced Hall-circuit motifs.

These results convert a large sparse completion problem into closed forms on defect-one trees and auditable low-dimensional programs on the remaining cores. They are algebraic and convex-analytic interface theorems. No estimate here proves that the literal fixed-shift arithmetic matrix has forest support, that its motif scores are non-negligible, or that any global endpoint budget is crossed.

Keywords: sparse completion; bipartite forest; matching rank; quotient gauge; one-defect formula; geometric median; dual packing.

Convention. The exact forest theorems concern complex-scalar coordinate blocks. Every physical specialization keeps one fixed nonzero shift h_0 , literal coefficients, complete native keys, actual nonzero support, and global raw atomic normalization. Graph zero means exact algebraic zero, never thresholded zero.

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1 Introduction

The completion layers developed in [4, 5] separate a fixed-shift physical assembly from a finite convex program. After immutable coordinates and exact zero columns are removed and the actual incidence graph is split into connected components, the active algebraic problem is

$$\Gamma(b, C) = \min_{\theta \in \mathbb{C}^J} \sum_{i \in I} \left| b_i + \sum_{j \in J} c_{ij} \theta_j \right|. \quad (1)$$

For an arbitrary matrix, the primal–dual certificate of TPC-77 is the right stopping point. The purpose of this paper is to identify what becomes exactly solvable when the *actual* nonzero support of C is a forest.

There are three distinct questions.

- (Q1) When does the parameter family cancel every datum?
- (Q2) If cancellation is impossible, how many independent obstructions remain and when is there a closed formula?
- (Q3) When the obstruction space is multidimensional, what exact reduction and lower certificates survive?

Forest support answers all three without genericity assumptions on the nonzero coefficients. Its decisive feature is not merely sparsity. Every matching minor contains a unique determinant monomial, so no coefficient cancellation can destroy its rank. This converts matching theory [1, 3] into an exact algebraic test.

Main results. Variable leaves can be removed exactly. Forest rank equals maximum matching size. Once variable-leaf pruning stops, a connected core has defect

$$d = \dim_{\mathbb{C}} \text{Ker } C^* = 1 + \sum_j (\deg(j) - 2).$$

The case $d = 1$ has a complete closed solution, including both optimizers and the entire optimal residual face. Higher defect is not hidden behind a scalar surrogate: it is reduced exactly to a d -complex-variable second-order cone problem. In parallel, a column-dominant pivot gives a different exact elimination rule, and balanced local charges give checkable dual-packing lower bounds.

What is genuinely new at this layer. Component decomposition alone says that independent pieces add. Here we determine the value of every defect-one forest piece in closed form and give a structural zero test for every forest piece. We also isolate the exact branching statistic that measures the dimension of the unsolved core. This is a real reduction of the completion interface, not a relabeling of the previous dual program.

Scope. The forest theorems use nonzero complex-scalar entries c_{ij} . TPC-77 allowed general real-linear maps $\mathbb{C} \rightarrow \mathbb{C}$; a nonzero real-linear leaf block need not be surjective, so those results do not extend verbatim. Moreover, forest topology says nothing about how frequently a literal fixed-shift arithmetic assembly realizes a given motif, or how large the corresponding datum score is. Those are later arithmetic-transfer questions. We keep them outside every theorem below.

Organization. Section 2 records the primal–dual normal form. Section 3 proves leaf erasure and the matching-rank theorem. Section 4 identifies the pruned-core defect. Sections 5 and 6 give the closed solution and phase transport. Section 7 proves a second exact pivot rule. Sections 8 and 9 treat branching cores and dual packings. Sections 10 and 11 state the failure modes and the physical claim boundary.

2 Complex-scalar completion and its support graph

Let I and J be finite sets, let $b \in \mathbb{C}^I$, and let $C = (c_{ij})_{i \in I, j \in J} \in \mathbb{C}^{I \times J}$. We use

$$\langle x, y \rangle = \sum_i \bar{x}_i y_i, \quad [x, y]_{\mathbb{R}} = \operatorname{Re} \langle x, y \rangle.$$

The component value is

$$\Gamma(b, C) = \min_{\theta \in \mathbb{C}^J} \|b + C\theta\|_1. \quad (2)$$

Since this is distance to a finite-dimensional subspace, the minimum is attained. Standard finite-dimensional convex duality [2] gives

$$\Gamma(b, C) = \max \left\{ [z, b]_{\mathbb{R}} : z \in \mathbb{C}^I, \|z\|_{\infty} \leq 1, C^* z = 0 \right\}. \quad (3)$$

Both extrema are attained. The condition $C^* z = 0$ is a complex linear balance equation because the variables in (2) range over $\mathbb{C}^J$.

Definition 2.1 (Actual incidence graph). The actual bipartite support graph is

$$G(C) = (I \sqcup J, E), \quad (i, j) \in E \iff c_{ij} \neq 0.$$

It is built after all algebraic sums defining C have been performed.

Permutation of rows and columns by the connected components of $G(C)$ block diagonalizes C . Therefore Γ is the sum of its connected component values, together with $|b_i|$ for isolated analysis rows. Isolated variable columns contribute nothing. We may accordingly prove the main statements for a connected component and sum them afterwards.

Remark 2.2 (Why scalar blocks are stated explicitly). In TPC-77 an edge could carry a general real-linear coordinate map $L : \mathbb{C} \rightarrow \mathbb{C}$. Here an edge carries multiplication by a nonzero complex number. This ensures that a leaf variable can realize every complex correction at its adjacent row and that a matching minor is an ordinary complex determinant. Section 10 shows that “nonzero real-linear block” is insufficient.

Definition 2.3 (Forest component and matching number). A component is a *forest component* when $G(C)$ is a tree. For a forest G , let $\nu(G)$ denote the largest number of pairwise vertex-disjoint edges.

We distinguish two types of leaves throughout: an analysis leaf belongs to I , while a variable leaf belongs to J . Only the second is unconditionally free.

3 Variable-leaf erasure and forest rank

Theorem 3.1 (Exact variable-leaf erasure). *Suppose $j \in J$ is adjacent to exactly one row $i \in I$, with $c_{ij} \neq 0$. Let C' and b' be obtained by deleting row i and column j . Then*

$$\Gamma(b, C) = \Gamma(b', C').$$

If deletion disconnects the remaining graph, the right-hand side is the sum of the resulting component values.

Proof. Fix all variables except θ_j . Because column j appears only in row i , choosing

$$\theta_j = -\frac{b_i + \sum_{k \neq j} c_{ik} \theta_k}{c_{ij}}$$

sets the i -th residual to zero and changes no other residual. Minimizing over the remaining variables proves the equality. Equivalently, the j -th dual balance equation is $\bar{c}_{ij} z_i = 0$, so every dual feasible vector has $z_i = 0$. $\square$

Definition 3.2 (Variable-leaf pruning). Repeatedly erase a variable leaf and its adjacent analysis row by [theorem 3.1](#); also erase every isolated variable produced by the operation. The surviving forest is called a *pruned core*.

Every erasure preserves the total value, irrespective of the order. Different orders may label the surviving components differently, but all yield exact residual programs.

Theorem 3.3 (Forest structural rank). *Let the actual support of C be a forest and suppose every supported entry is nonzero. Then*

$$\text{rank}_\mathbb{C} C = \nu(G(C)). \quad (4)$$

Proof. For every matrix, a nonzero $r \times r$ minor supplies a size- r matching through one nonzero determinant term. Hence $\text{rank } C \leq \nu(G(C))$.

Conversely, take a matching of size r and the submatrix on its incident rows and columns. The matching gives one nonzero determinant monomial. If another determinant monomial were nonzero, the two perfect matchings of this induced bipartite graph would have a nonempty alternating cycle in their symmetric difference. That is impossible in a forest. The determinant is therefore, up to sign, the product of the matched nonzero entries and is nonzero. Thus $\text{rank } C \geq r$. Maximize over matchings. $\square$

Corollary 3.4 (Uniform zero-cost test). *For a forest-supported C , the following are equivalent:*

- (i) $\Gamma(b, C) = 0$ for every $b \in \mathbb{C}^I$;
- (ii) C has full row rank;
- (iii) $G(C)$ has a matching saturating every analysis row;
- (iv) variable-leaf pruning erases every analysis row.

Proof. The first two statements are equivalent to $\text{Ran } C = \mathbb{C}^I$, and [theorem 3.3](#) gives the equivalence with (iii).

If pruning erases all rows, repeated use of [theorem 3.1](#) gives zero value for every datum, hence (iv) implies (i). Conversely, if a nonempty pruned core remains, every surviving variable has degree at least two. On a connected core with p rows and q variables, the tree identity gives

$$p + q - 1 = |E| \geq 2q,$$

so $p \geq q + 1$. Such a core has no row-saturating matching. Hence (iii) implies (iv): indeed, when a variable leaf j adjacent to i is erased, any row-saturating matching can first be modified to use (i, j) (freeing the edge formerly incident to i), and then restricted to the remaining graph. Row saturation is therefore preserved at every pruning step. $\square$

Remark 3.5 (Datum-specific cancellation). The graph test is uniform in b . For one particular datum,

$$\Gamma(b, C) = 0 \iff -b \in \text{Ran } C,$$

even when no row-saturating matching exists. Topology alone cannot decide this datum-specific equality.

4 The exact defect of a pruned tree core

Let a nontrivial connected pruned core contain p analysis rows and q variables. Every variable now has degree at least two.

Lemma 4.1 (All surviving variables are matchable). *Every set $S \subseteq J$ of variables in a pruned tree core satisfies*

$$|N(S)| \geq |S| + 1.$$

In particular, a matching saturates every variable.

Proof. The subgraph induced by $S \cup N(S)$ is a forest. Because every edge incident to a variable of S remains inside this induced graph and every such variable has degree at least two,

$$2|S| \leq |E(S, N(S))| \leq |S| + |N(S)| - 1.$$

The stated inequality follows. Hall's theorem then supplies a matching saturating J . $\square$

Theorem 4.2 (Pruned-core defect identity). *For a connected pruned tree core,*

$$\text{rank}_{\mathbb{C}} C = q, \quad d := \dim_{\mathbb{C}} \text{Ker } C^* = p - q = 1 + \sum_{j \in J} (\deg(j) - 2). \quad (5)$$

Proof. Lemma 4.1 and theorem 3.3 give $\text{rank } C = q$, hence $\dim \text{Ker } C^* = p - q$. The tree identity

$$p + q - 1 = |E| = \sum_{j \in J} \deg(j)$$

rearranges to the last equality in (5). $\square$

Corollary 4.3 (One-defect topology). *A nontrivial pruned tree core has $d = 1$ if and only if every variable has degree two. Each unit of excess degree $\deg(j) - 2$ adds one complex dual degree of freedom.*

Thus branching is measured exactly. A degree-three variable is not a minor perturbation of the scalar case: it creates at least a two-complex-dimensional dual feasible section. An isolated analysis row is treated separately as the degenerate $p = 1, q = 0$ core and has value $|b_i|$.

Remark 4.4 (Combinatorics versus coefficients). The dimension formula is coefficient-independent only because forest support prevents determinant cancellation. It does not say that the value $\Gamma(b, C)$ is coefficient-independent. Even at defect one, the transported charge and its pairing with b retain all literal edge magnitudes and phases.

5 Closed form at one complex defect

The next theorem does not require forest support; it applies whenever the dual obstruction space is one complex dimensional.

Theorem 5.1 (One-defect primal–dual formula). *Assume*

$$\text{Ker } C^* = \mathbb{C}\alpha, \quad \alpha \neq 0.$$

Set

$$\beta = \langle \alpha, b \rangle, \quad M = \|\alpha\|_{\infty}.$$

Then

$$\boxed{\Gamma(b, C) = \frac{|\beta|}{M}}. \quad (6)$$

If $\beta = 0$, the unique optimal residual is $y = 0$, and $z = 0$ is a dual optimizer. If $\beta \neq 0$, a dual optimizer is

$$z_* = \frac{\beta}{M|\beta|} \alpha. \quad (7)$$

For any r with $|\alpha_r| = M$, define

$$y_r = \frac{\beta}{\alpha_r}, \quad y_i = 0 \quad (i \neq r). \quad (8)$$

Then $y - b \in \text{Ran } C$, so any solution of $C\theta = y - b$ is a primal optimizer.

Proof. Every dual feasible charge has the form $z = \lambda\alpha$ with $|\lambda|M \leq 1$. Therefore (3) becomes

$$\max_{|\lambda| \leq M^{-1}} \operatorname{Re}(\bar{\lambda}\beta) = \frac{|\beta|}{M},$$

attained by $\lambda = \beta/(M|\beta|)$ when $\beta \neq 0$.

For (8),

$$\langle \alpha, y \rangle = \bar{\alpha}_r \frac{\beta}{\alpha_r} = \beta = \langle \alpha, b \rangle.$$

Thus $y - b \perp \operatorname{Ker} C^*$. In finite dimensions, $(\operatorname{Ker} C^*)^\perp = \operatorname{Ran} C$, so $C\theta = y - b$ is solvable. Finally, $\|y\|_1 = |\beta|/M$, matching the dual value. $\square$

The proof gives more than one optimizer.

Theorem 5.2 (Complete optimal-residual face). *Assume the hypotheses of theorem 5.1 and $\beta \neq 0$. Let*

$$S = \{i \in I : |\alpha_i| = M\}.$$

A residual $y \in b + \operatorname{Ran} C$ minimizes $\|y\|_1$ if and only if

$$y_i = 0 \ (i \notin S), \quad y_i = t_i(z_*)_i \ (i \in S), \quad t_i \geq 0, \quad \sum_{i \in S} t_i = \frac{|\beta|}{M}. \quad (9)$$

In particular, a unique maximum-modulus coordinate of α forces a unique optimal residual.

Proof. For any feasible residual,

$$\|y\|_1 \geq [z_*, y]_{\mathbb{R}} = [z_*, b]_{\mathbb{R}} = \Gamma(b, C).$$

Equality in the coordinatewise inequalities $\operatorname{Re}(\overline{(z_*)_i} y_i) \leq |(z_*)_i| |y_i| \leq |y_i|$ holds exactly when $y_i = 0$ wherever $|(z_*)_i| < 1$, and $y_i = t_i(z_*)_i$ with $t_i \geq 0$ wherever $|(z_*)_i| = 1$. These are precisely the support and phase conditions in (9). They satisfy $\langle \alpha, y \rangle = \beta$ exactly when the displayed sum of the t_i is $|\beta|/M$. Hence they are also sufficient. $\square$

Corollary 5.3 (Exact zero and positive obstruction). *Under the one-defect hypothesis,*

$$\Gamma(b, C) = 0 \quad \Longleftrightarrow \quad \langle \alpha, b \rangle = 0.$$

Thus one literal complex scalar is the complete obstruction.

6 Degree-two trees and phase transport

Let a pruned tree core have only degree-two variables. Contract each variable j , adjacent to rows u and v , into an edge between those rows. The contracted graph is a tree on I .

Theorem 6.1 (Explicit transported obstruction). *Choose a root row r and set $\alpha_r = 1$. Orient every contracted edge away from the root. If variable j connects parent row u to child row v , define*

$$\alpha_v = -\frac{\bar{c}_{uj}}{c_{vj}} \alpha_u. \quad (10)$$

Then every α_i is nonzero,

$$C^* \alpha = 0, \quad \operatorname{Ker} C^* = \mathbb{C} \alpha,$$

and the value is

$$\Gamma(b, C) = \frac{|\sum_{i \in I} \bar{\alpha}_i b_i|}{\max_{i \in I} |\alpha_i|}. \quad (11)$$

Proof. There is a unique root path to every row, so (10) is consistent and nonzero. For the column j joining u, v , it gives

$$\overline{c_{uj}}\alpha_u + \overline{c_{vj}}\alpha_v = 0.$$

Thus $C^*\alpha = 0$. The core has one complex defect by corollary 4.3, so α spans the kernel. Theorem 5.1 gives (11). $\square$

Proposition 6.2 (Recovery of primal variables). *Let y be any optimal residual supplied by theorem 5.2 and set $d = y - b$. The solution of $C\theta = d$ can be recovered by repeatedly removing a nonroot analysis leaf ℓ . If j is its incident variable and u is the parent row, set*

$$\theta_j = \frac{d_\ell}{c_{\ell j}}, \quad d_u \leftarrow d_u - c_{uj}\theta_j,$$

then delete ℓ, j . The last root equation holds automatically.

Proof. Each step solves the leaf equation exactly and substitutes it into its parent equation. After all nonroot rows are removed, the only possible remaining inconsistency is in the root row. But $\langle \alpha, d \rangle = 0$ and the transported nonzero coefficients express the root equation as the unique linear dependence among the rows, so it also vanishes. $\square$

Corollary 6.3 (Weighted incidence columns). *Suppose every contracted edge $e = \{u, v\}$ has column*

$$C_e = w_e(e_u - e_v), \quad w_e \neq 0.$$

Then $\alpha \equiv 1$ and

$$\Gamma(b, C) = \left| \sum_{i \in I} b_i \right|. \quad (12)$$

Equation (12) is the unique conservation law of a tree incidence matrix. For general complex edge coefficients, the ordinary sum is replaced by the literal phase-and-magnitude transported sum in (11).

Remark 6.4 (No cycle holonomy on a tree). The recursion (10) never compares two paths to the same row. A cycle would impose a multiplicative consistency condition on the transported phases and magnitudes. That genuinely new phenomenon is not part of the forest theorem.

7 A column-dominant exact pivot

Variable leaves are maximally sparse pivots. A larger class of exact ℓ^1 eliminations is available when one coefficient dominates the rest of its column.

Theorem 7.1 (Column-dominant pivot elimination). *Choose a row $r \in I$ and variable $j \in J$ with $c_{rj} \neq 0$. Assume*

$$|c_{rj}| \geq \sum_{i \neq r} |c_{ij}|. \quad (13)$$

For $i \neq r$ and $k \neq j$, define

$$b'_i = b_i - \frac{c_{ij}}{c_{rj}}b_r, \quad c'_{ik} = c_{ik} - \frac{c_{ij}}{c_{rj}}c_{rk}. \quad (14)$$

Then

$$\Gamma(b, C) = \Gamma(b', C'), \quad (15)$$

where C' deletes row r and column j after the update. There exists an optimizer whose r -th residual is zero. If (13) is strict, every optimizer has zero r -th residual.

Proof. Fix all variables $k \neq j$, and let

$$u_i = b_i + \sum_{k \neq j} c_{ik}\theta_k.$$

Write $\theta_j = -u_r/c_{rj} + \delta$. The objective becomes

$$|c_{rj}\delta| + \sum_{i \neq r} \left| u_i - \frac{c_{ij}}{c_{rj}} u_r + c_{ij}\delta \right|.$$

The triangle inequality bounds this below by its value at $\delta = 0$ plus

$$\left(|c_{rj}| - \sum_{i \neq r} |c_{ij}| \right) |\delta|.$$

Thus $\delta = 0$ is always optimal under (13), and is forced under strict dominance. Substitution at $\delta = 0$ produces (14) and (15). $\square$

Proposition 7.2 (Matched dual lift). *Under (13), every dual feasible z' for the reduced program lifts by*

$$z_i = z'_i \ (i \neq r), \quad z_r = -\frac{1}{c_{rj}} \sum_{i \neq r} \overline{c_{ij}} z'_i. \quad (16)$$

The lift satisfies $|z_r| \leq 1$, $C^* z = 0$, and $[z, b]_{\mathbb{R}} = [z', b']_{\mathbb{R}}$.

Proof. The bound follows from $\|z'\|_{\infty} \leq 1$ and (13). Formula (16) is exactly the j -column balance equation. Substitution shows that every other column balance is the corresponding reduced equation, while the objective identity follows from the update of b' . $\square$

A component that can be exhausted by successive pivots satisfying (13) is *dominance-peelable*. The theorem and its dual lift then give an exact recursion with matched upper and lower certificates. A variable leaf is the special case in which every off-pivot entry of column j is zero. An analysis leaf is not such a case unless its incident column also satisfies the dominance inequality.

8 Exact reduction of a branching core

Let a connected pruned tree core have p rows, q variables, and defect $d = p - q \geq 1$. By lemma 4.1, choose a matching that saturates all variables. Let $M \subset I$ be its q matched rows and $U = I \setminus M$ the d unmatched rows. Write C_M and C_U for the corresponding row restrictions.

Theorem 8.1 (Defect-dimensional dual reduction). *The square matrix C_M is invertible. Define*

$$K : \mathbb{C}^U \longrightarrow \mathbb{C}^I, \quad (K\xi)_U = \xi, \quad (K\xi)_M = -(C_M^*)^{-1} C_U^* \xi.$$

Then K is an isomorphism from $\mathbb{C}^U$ onto $\text{Ker } C^$, and*

$$\Gamma(b, C) = \max_{\xi \in \mathbb{C}^U} \{ [K\xi, b]_{\mathbb{R}} : \|K\xi\|_{\infty} \leq 1 \}. \quad (17)$$

Thus a branching tree core reduces exactly to a complex second-order-cone program in d complex variables.

Proof. The matching is the unique perfect matching of the forest subgraph on $M \sqcup J$; otherwise the symmetric difference of two perfect matchings would contain a cycle. Hence $\det C_M$ is, up to sign, the product of its matched nonzero entries. It is nonzero.

Writing $z = (z_M, z_U)$, the balance equation is

$$C_M^* z_M + C_U^* z_U = 0.$$

After setting $z_U = \xi$, invertibility gives the displayed formula for z_M , uniquely. Substitution into (3) proves (17). Each constraint $|(K\xi)_i| \leq 1$ is a two-dimensional second-order-cone constraint. $\square$

Remark 8.2 (What the reduction does and does not close). At $d = 1$, (17) is a disk and reduces to theorem 5.1. At $d > 1$, it is an intersection of complex ellipsoidal cylinders. The theorem removes all redundant primal variables and exposes the exact intrinsic dimension, but it does not replace a genuinely multidimensional feasible body by a fictitious single scalar.

Proposition 8.3 (Exact a posteriori certificate). *For any primal trial θ and reduced dual trial ξ with $\|K\xi\|_\infty \leq 1$, set*

$$U(\theta) = \|b + C\theta\|_1, \quad L(\xi) = [K\xi, b]_{\mathbb{R}}.$$

Then

$$L(\xi) \leq \Gamma(b, C) \leq U(\theta), \quad 0 \leq U(\theta) - L(\xi).$$

Equality is a complete optimality certificate.

This certificate is stable under the component sum of TPC-77: upper and lower values add, so the global gap is the sum of component gaps.

Remark 8.4 (Exact min-sum remains available). Rooting a general tree also yields an exact convex min-sum recursion. For real data and real scalar variables the messages are piecewise-linear. For genuinely complex data they are generally curved cone-representable functions. We use (17) as the canonical finite certificate because it does not suggest a nonexistent coordinatewise-median formula.

9 Hall-circuit motifs and dual packings

The reduced program is exact, but a lower bound may be certified without solving it globally. The correct local objects are balanced charges, not merely dense-looking subgraphs.

Definition 9.1 (Row Hall circuit). A nonempty row set $S \subseteq I$ is a *row Hall circuit* when

$$|N(S)| < |S|$$

and no proper nonempty subset of S has this property. Here $N(S)$ contains *every* variable adjacent to a row of S .

Theorem 9.2 (Canonical circuit charge). *Assume $G(C)$ is a forest and S is a row Hall circuit. Then*

$$|N(S)| = |S| - 1, \quad \dim_{\mathbb{C}} \text{Ker } C_{S, N(S)}^* = 1.$$

*Every nonzero generator g^S of this kernel has $g_i^S \neq 0$ for all $i \in S$. Extending g^S by zero outside S gives a global balanced charge $C^*g^S = 0$. Consequently*

$$\Gamma(b, C) \geq \frac{|\langle g^S, b_S \rangle|}{\|g^S\|_\infty}. \quad (18)$$

Proof. For each $i \in S$, minimality gives $|N(S \setminus \{i\})| \geq |S| - 1$, while $N(S \setminus \{i\}) \subseteq N(S)$. Hence $|N(S)| = |S| - 1$.

Every proper row subset is matchable, whereas S is not. The forest-rank theorem therefore gives row rank $|S| - 1$ for $C_{S, N(S)}$, so its left kernel is one dimensional. If a generator vanished at i , its restriction would give a nonzero left-kernel vector on the proper set $S \setminus \{i\}$, contradicting its full row rank. The zero extension is globally balanced because $N(S)$ contains all columns incident to S . Normalize it to ℓ^∞ -norm one, rotate its phase to make the pairing with b nonnegative, and apply (3). $\square$

The insistence on the full neighbor set is essential. A charge balanced only against columns drawn inside a picture of the motif may fail the global equation at an incident boundary column.

Theorem 9.3 (Disjoint and fractional motif packing). *Let $g^1, \dots, g^m \in \text{Ker } C^*$.*

(a) If their row supports are pairwise disjoint and $\|g^s\|_\infty = 1$, then

$$\Gamma(b, C) \geq \sum_{s=1}^m |\langle g^s, b \rangle|. \quad (19)$$

(b) More generally, if

$$\sum_{s=1}^m |g_i^s| \leq 1 \quad (i \in I), \quad (20)$$

then the same lower bound (19) holds.

Proof. For each s , choose $\omega_s \in \mathbb{C}$, $|\omega_s| = 1$, so that $\langle \omega_s g^s, b \rangle$ is nonnegative real. The sum

$$z = \sum_s \omega_s g^s$$

is balanced. Disjoint support in (a), or the capacity inequality in (b), gives $\|z\|_\infty \leq 1$. Its dual objective is the stated sum. $\square$

Remark 9.4 (Certificate, not prevalence). [Theorem 9.3](#) is useful only after actual balanced motifs and their literal datum scores have been exhibited. Counting an ambient graph motif, replacing an exact zero by a tolerance, or ignoring incident columns does not produce a feasible charge. Moreover, a large abstract packing has arithmetic force only if its global raw normalization is kept and its physical carrier set has the required size.

10 Sharp countermodels and stop rules

The following examples prevent five tempting but false extensions.

Example 10.1 (An analysis leaf is not free). *Let*

$$C = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both analysis rows are leaves, but

$$\min_{\theta \in \mathbb{C}} (|1 + \theta| + |1 - \theta|) = 2.$$

For $b = (1, -1)^\top$, the value is instead zero. The same tree topology therefore implies neither a positive cost nor a saving for a specific datum.

Example 10.2 (A complex star is a geometric median). *For one variable with points $0, 1, i$, the objective is*

$$|\theta| + |\theta - 1| + |\theta - i|.$$

The coordinatewise median is 0 and has value 2. The planar Fermat point

$$\theta_* = \frac{3 - \sqrt{3}}{6}(1 + i)$$

has value $\sqrt{2 + \sqrt{3}} < 2$. Complex star components are weighted geometric-median problems; only collinear data reduce to an ordinary weighted median. With two points p_1, p_2 and weights w_1, w_2 , the exact value is

$$\min(w_1, w_2)|p_1 - p_2|.$$

Example 10.3 (Cycle support does not fix rank). *The matrices*

$$C_+ = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad C_- = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

have the same $K_{2,2}$ support, but ranks one and two. For $b = (1, -1)^\top$,

$$\Gamma(b, C_+) = 2, \quad \Gamma(b, C_-) = 0.$$

The forest hypothesis in [theorem 3.3](#) is therefore necessary.

Example 10.4 (Nonzero real-linear leaves need not be surjective). *Let a single real-linear block be $C(z) = \operatorname{Re} z$, regarded as a map $\mathbb{C} \rightarrow \mathbb{C}$, and take $b = i$. Then*

$$\min_{z \in \mathbb{C}} |i + \operatorname{Re} z| = 1.$$

The one-edge graph is a tree, but the leaf cannot be erased. General real-linear blocks require a surjective leaf map; the matching and phase-transport formulas require further invertibility and conformality.

Example 10.5 (Real and complex parameter fields differ).

$$\min_{t \in \mathbb{R}} |i + t| = 1, \quad \min_{t \in \mathbb{C}} |i + t| = 0.$$

For real parameters the dual stationarity condition is a real projection equation, not $C^*z = 0$ as a complex equation. No theorem may switch parameter fields silently.

Remark 10.6 (Topology is not a lower bound by itself). A Hall circuit supplies a one-dimensional space of possible charges. Its score can still vanish because $\langle g^S, b_S \rangle = 0$. Likewise, many motifs can overlap so strongly that their fractional packing has small total capacity. Arithmetic usefulness needs both carrier prevalence and noncancelling literal scores.

11 Literal fixed-shift census and claim boundary

The exact theorems become a physical certificate only after a literal assembly has been archived. Fix one nonzero physical shift h_0 . For each truncation or scale X , the required record is:

- (P1) the native input and output keys, with no collapsed or missing carrier labels;
- (P2) the literal baseline vector b_X and completion matrix C_X ;
- (P3) the exact nonzero support after all algebraic sums;
- (P4) the decomposition into fixed rows, zero columns, and actual connected components;
- (P5) for every forest component, its matching, pruning trace, defect, transported charge or reduced matrix K ;
- (P6) matched primal and dual values, with numerical tolerances recorded separately from graph adjacency;
- (P7) the common global raw atomic normalizer, whether it is already absorbed into b_X, C_X , and every component contribution under that single convention.

Proposition 11.1 (Auditable forest census identity). *For any archived finite literal program, let $\mathcal{F}_X$ be its actual forest components and $\mathcal{U}_X$ its remaining components. Then*

$$\Gamma(b_X, C_X) = \|b_{X, \text{fix}}\|_1 + \sum_{S \in \mathcal{F}_X} \Gamma_S + \sum_{S \in \mathcal{U}_X} \Gamma_S. \quad (21)$$

Every zero-prunable forest term and every defect-one forest term in (21) is exactly computable by this paper. Every other forest term admits the defect-reduced interval of this paper. Every remaining term, including a nonforest component, retains the matched TPC-77 primal–dual interval.

Proof. This is the exact component factorization of TPC-77, followed by [theorems 3.1, 5.1 and 8.1](#) and [corollary 3.4](#) on the forest terms. No thresholding or change of normalization occurs. $\square$

No literal fixed- h_0 dataset is included in this paper. Therefore we make no numerical statement about the frequency of forest components, the distribution of their defects, or the normalized size of their scores.

Progress labels

Label	Status after this paper
L0: model algebra	Exact for the finite complex-scalar forest program.
L1: interface/certificate	Advanced: forest rank, pruning, defect-one values, reduced SOCPs, and motif packings are now exact and auditable.
L2: fixed- h_0 arithmetic	Not advanced here: no density, cancellation, or endpoint estimate for literal arithmetic coefficients is proved.

Endpoint loss ledger and stop rule

The inherited arithmetic ledger permits a *total fixed-power exponent loss strictly smaller than* $1/400$. Equality is not enough: the route stops once accumulated certified losses reach or exceed $1/400$. The exact finite-dimensional reductions in this paper add zero exponent loss by themselves, but they also prove no arithmetic gain. Any downstream dictionary, carrier restriction, approximation, or reassembly loss must be entered in the same ledger.

In particular, neither a component gauge nor a motif-packing lower score is a $1/400$ gain without a separate literal transfer theorem that proves the direction, scale, and global raw normalization. If that transfer incurs total loss at least $1/400$, or reduces to an unproved parity-sensitive correlation, this branch must be recorded as an L1 interface result or a negative transfer result rather than an L2 advance.

Conclusion

Forest sparsity creates an exact hierarchy:

variable-leaf pruning $\longrightarrow$ uniform zero test,
degree-two core $\longrightarrow$ one-obstruction closed form,
branching core $\longrightarrow$ defect-dimensional SOCP,
balanced motifs $\longrightarrow$ additive dual lower certificates.

This closes a substantial finite-dimensional subgate of sparse provenance completion. It does not identify the needed physical carriers or prove a parity-breaking arithmetic estimate. That precise separation is the basis for the next transfer layer.

References

- [1] László Lovász and Michael D. Plummer. *Matching Theory*. North-Holland, 1986.
- [2] R. Tyrrell Rockafellar. *Convex Analysis*. Princeton University Press, 1970. doi: 10.1515/9781400873173.
- [3] Alexander Schrijver. *Combinatorial Optimization: Polyhedra and Efficiency*. Springer, 2003.
- [4] Liang Wang. Primal–dual completion certificates for fixed-shift primitive entry kernels: Exact quotient gauges, obstruction witnesses, and perron aggregation, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-75-primal-dual-completion-certificates. Preprint.
- [5] Liang Wang. Exact sparse incidence decomposition for provenance completion programs: Componentwise primal–dual certificates and verified carrier census protocols, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-77-sparse-incidence-completion-certificates. Preprint.